

Predicting Waitlist Risk and Time-to-Transplant for Kidney Candidates

A reproducible, fairness-aware analysis of national OPTN registry data — who is at risk, how long they wait, and whether the model is fair and explainable.

Cohort 2015+ kidney-alone waitlist **Data** OPTN STAR KIDPAN · 202606 **Tasks** classification + survival
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494,862 kidney-alone registrations 2015+ cohort	14.1% adverse-outcome rate died / too-sick	0.726 ROC-AUC calibrated XGBoost	0.212 → 0.111 Brier score after isotonic calibration
0.629 C-index time-to-transplant	2.51× top-decile lift risk concentration		

Abstract

Background. Patients on the U.S. kidney transplant waiting list face two intertwined uncertainties: whether they will survive to transplant, and how long they will wait. With more than 90,000 candidates awaiting a kidney but fewer than 30,000 transplants performed annually [1], the supply–demand gap makes waitlist risk stratification clinically consequential. Outcomes are shaped both by patient factors and by a national allocation system that changed materially in 2014 (Kidney Allocation System, KAS [4]) and 2021 (Acuity Circles [5]). We build and audit predictive models for both questions

using national registry data, with an explicit emphasis on leakage control, probability calibration, fairness across demographic and geographic subgroups, and reproducibility.

Methods. From the OPTN **KIDPAN** Standard Transplant Analysis and Research (STAR) file we constructed a cohort of **494,862 kidney-alone waitlist registrations** listed on or after 1 January 2015. We framed two tasks: (1) a **binary classifier** for an adverse waitlist outcome (death or removal-as-too-sick before transplant; 14.1% prevalence) using **listing-time features only**, and (2) a **time-to-transplant survival model** (Cox proportional hazards) treating death, removal, and administrative censoring as right-censoring. Models were a logistic-regression baseline and a gradient-boosted tree ensemble (XGBoost). We applied **post-hoc isotonic calibration**, evaluated with ROC-AUC, PR-AUC, Brier score, top-decile lift, and Harrell's C-index, audited subgroup AUC against a 0.05 fairness tolerance, and explained the model with SHAP.

Results. The calibrated XGBoost classifier achieved ROC-AUC 0.726, PR-AUC 0.284, top-decile lift $2.51\times$, and — after isotonic calibration — a Brier score of **0.111** (down from 0.212 uncalibrated), meeting the Brier and lift targets while falling short of the aspirational AUC/PR-AUC targets. The Cox model reached a C-index of 0.629 for time-to-transplant, with median observed wait ranging **211–400 days across OPTN regions**. Subgroup AUC gaps stayed within the 0.05 tolerance for sex (0.004), race/ethnicity (0.024), blood type (0.044), region (0.034), and age (0.042). SHAP identified **functional status at listing, policy era, BMI, ethnicity, and region** as the leading drivers.

Conclusion. With only listing-time registry features, waitlist-risk and wait-time are predictable to a clinically suggestive but not decisive degree. The most consequential methodological findings are that (a) a naïve cohort definition silently collapses the waitlist to transplant recipients, (b) class-imbalance weighting must be paired with calibration before probabilities are trusted, and (c) subgroup parity in AUC coexists with large *absolute* differences in wait time — a fairness signal that ranking metrics alone conceal.

Keywords: organ allocation, kidney transplantation, survival analysis, competing risks, calibration, algorithmic fairness, SHAP, OPTN/UNOS, electronic registry.

Plain-Language Summary

What we did. We used national records on people waiting for a kidney transplant to build two prediction tools: one that estimates a patient's chance of dying or becoming too sick before a kidney becomes available, and one that estimates how long they will wait.

Why it matters. Far more people need a kidney than there are kidneys to give. Many people wait years, and some die waiting. A tool that flags who is most at risk could help care teams focus attention where it is needed most.

What we found. Using only information known when a patient joins the list, the model correctly ranks a higher-risk patient above a lower-risk one about **73% of the time**, and its highest-risk 10% of patients have adverse outcomes at about **2.5× the average rate**. After a correction step, the risk percentages it reports are trustworthy (well-calibrated). Predicting *wait time* was harder.

The catch. The model performs about equally well for men and women, across races, and across regions — but that hides a real problem: how *long* people wait varies almost two-fold depending on which region they live in. Fair-looking accuracy is not the same as fair access.

Bottom line. Waiting-list risk can be predicted moderately well from basic listing information — useful for prioritizing care, but not precise enough to decide who gets an organ, and never a substitute for clinical judgment.

A note on jargon. Throughout, *waitlist* = the list of people waiting for a transplant; *censored* = a patient still waiting when we stopped observing, so we don't yet know their final outcome; *calibration* = whether a predicted "20% risk"

really happens about 20% of the time; *AUC* = how well the model ranks a riskier patient above a safer one (0.5 = coin-flip, 1.0 = perfect); *SHAP* = a method that shows which patient details pushed a prediction up or down.

1. Introduction

1.1 CLINICAL PROBLEM

End-stage kidney disease is treatable by transplantation, which confers a survival and quality-of-life advantage over long-term dialysis. Demand, however, vastly exceeds supply. As of the 2024 SRTR Annual Data Report, more than 90,000 candidates are awaiting a kidney and new listings reached a record in 2024, while fewer than 30,000 kidney transplants are performed annually (27,759 in 2024) [1], [2]. The consequence is a long, hazardous wait: many candidates die or deteriorate to the point of being removed as "too sick to transplant" before an organ is offered. Pretransplant mortality varies widely by geography — on the order of 2.5–7.0 deaths per 100 patient-years across donation service areas [1] — and a reliable *median* waiting time can no longer be computed because fewer than half of listed candidates are transplanted in a given year; historically it approached four years [1].

Two operational questions follow directly for clinicians and candidates:

1. **Who is at risk?** Which candidates are most likely to die or be removed before transplant?
2. **How long will they wait?** What is the expected time to transplant, accounting for candidates still waiting when observation ends?

A third question governs whether any resulting model should be trusted or deployed:

3. **Is the model fair and explainable?** Does performance hold across demographic and geographic subgroups, and can its drivers be stated in plain clinical language?

1.2 WHY THE PROBLEM IS HARD

This problem is harder than it looks, for five reasons. **(1) The two questions are linked** — how likely someone is to die on the list depends on how long they have to wait. **(2) The rules keep changing.** Two national policy reforms — KAS in 2014 [4] and Acuity Circles in 2021 [5] — changed how kidneys are handed out, so a model trained across those years is aiming at a moving target. **(3) Many outcomes are still unknown.** A large share of patients are still waiting when the data are pulled (statisticians call this *censored*), and transplant, death, and removal compete with one another — a patient who dies can never be transplanted [8]. **(4) The bad event is rare.** Far fewer people have a bad outcome than get transplanted, so plain "accuracy" looks deceptively good. **(5) The biggest trap:** a model can score well simply by memorizing who *historically* got transplants — including the system's racial and geographic unfairness [9], [10], [11] — instead of learning real medical risk. For these reasons we treat data leakage, censoring, competing risks, calibration, and fairness as core design choices rather than afterthoughts.

1.3 CONTRIBUTIONS

- A **carefully defined kidney-alone waitlist cohort** from the OPTN [KIDPAN](#) Standard Transplant Analysis and Research (STAR) file, with the cohort-construction pitfalls documented (Section 3.3) so they are not silently repeated.
- A **leakage-controlled** classification model using only variables known at listing, and an explicit enumeration of the transplant-record fields that must be excluded.
- A demonstration that **post-hoc isotonic calibration** repairs the Brier-score damage caused by class-imbalance weighting ($0.212 \rightarrow 0.111$) without altering

ranking performance, echoing calibration's status as the "Achilles heel" of clinical prediction [12], [13].

- A **fairness audit** showing that subgroup-AUC parity within 0.05 can coexist with large absolute wait-time disparities, and a discussion of why that distinction matters for equity [9], [10].
- Full **reproducibility artifacts**: extract script, analysis pipeline, and aggregate-only outputs compatible with the governing Data Use Agreement.

2. Background and Related Work

2.1 KIDNEY ALLOCATION AND POLICY SHIFTS

U.S. deceased-donor kidney allocation is administered by the Organ Procurement and Transplantation Network (OPTN) under contract to HRSA, operated by UNOS. Two reforms bound our study period. The **Kidney Allocation System (KAS)**, effective 4 December 2014, replaced a largely first-come/first-served scheme with longevity matching (top-20% KDPI kidneys directed to top-20% EPTS candidates), backdating of waiting time to dialysis initiation, and additional priority for highly sensitized (high-CPRA) and blood-type-B candidates [4].

Acuity Circles, effective 15 March 2021, eliminated Donation Service Area and OPTN region as units of distribution, replacing them with fixed-distance concentric circles (first unit: a 250-nautical-mile radius around the donor hospital), with the goal of reducing geographic inequity [5]. Because allocation logic changes the feature-to-outcome mapping, a model spanning these boundaries risks **distribution shift**; we therefore encode policy era as a feature and analyze it directly.

2.2 PREDICTIVE MODELING OF TRANSPLANT OUTCOMES

Machine learning has been applied extensively to transplant outcomes, though more often to post-transplant graft/patient survival than to waitlist risk.

Senanayake *et al.* [7] systematically reviewed 18 predictive models for kidney graft failure, finding tree-based methods most common but noting that only one modeled time-to-event and that heterogeneity limited comparability. For the waitlist specifically, Salehinejad *et al.* [6] used an ensemble of random forests, XGBoost, and extra-trees to predict kidney-waitlist mortality, reporting that adding an abdominal-arterial-calcification score raised accuracy from 68% to 78% — evidence both that ML is applicable here and that listing-time features alone leave substantial residual uncertainty. Our study differs from prior work in three respects: strict listing-time feature hygiene to prevent leakage, calibration as a first-class deliverable, and an explicit subgroup-fairness audit.

2.3 SURVIVAL ANALYSIS AND COMPETING RISKS

The Scientific Registry of Transplant Recipients (SRTR) publishes national outcome reports [1], and the statistical literature has repeatedly cautioned that waitlist analyses must respect competing risks. Sapir-Pichhadze *et al.* [8] provide a tutorial specifically for waitlisted kidney candidates, contrasting cause-specific hazard models with the Fine–Gray subdistribution model [15] and showing that the choice materially changes conclusions. We adopt a cause-specific Cox model [14] for time-to-transplant and flag Fine–Gray as the recommended extension (Section 6).

2.4 FAIRNESS AND EQUITY IN KIDNEY TRANSPLANTATION

Access to kidney transplantation is known to be inequitable along racial lines. Ku *et al.* [10] showed that the race-adjusted eGFR equation delayed Black patients' eligibility for preemptive waitlisting, and Inker *et al.* [11] subsequently developed race-free eGFR equations now widely adopted. More broadly, Obermeyer *et al.* [9] demonstrated that a widely used clinical risk algorithm encoded substantial racial bias because it optimized a biased proxy (cost) rather than illness. These findings

motivate our central fairness concern: a waitlist-risk model that predicts well may simply be reproducing an inequitable allocation history, and subgroup *ranking* parity does not guarantee *outcome* equity.

2.5 CALIBRATION AND EXPLAINABILITY

Van Calster *et al.* [12], [13] argue that calibration — the agreement between predicted probabilities and observed frequencies — is the most neglected yet decision-critical property of clinical prediction models. We treat it as a required deliverable and use isotonic regression to repair the miscalibration induced by imbalance weighting. For interpretability we use SHAP [17], a game-theoretic attribution method with a consistent additive-feature-importance guarantee, computed via XGBoost's [16] native contribution output.

3. Data

3.1 SOURCE

The OPTN STAR files are the de-identified national research extracts of the transplant registry, governed by a Data Use Agreement (DUA). We used the combined kidney–pancreas file, `KIDPAN_DATA.DAT`, from the June 2026 release: **1,303,788 records × 475 variables**, delivered as tab-delimited text with an accompanying SAS-format data dictionary (`KIDPAN_DATA.htm`). The registry spans registrations back to 1987; our working scope is deliberately narrower (Section 3.2) to keep preprocessing tractable and the cohort clinically coherent.

3.2 COHORT DEFINITION

We restricted to **kidney-alone waitlist registrations listed on or after 1 January 2015**:

- **Organ:** waitlisted-organ field `WL_ORG == "KI"` . This excludes simultaneous kidney-pancreas (`KP`), pancreas-alone (`PA`), and islet (`PI`) registrations.
- **Listing window:** `INIT_DATE >= 2015-01-01` , keeping the cohort within the post-KAS era and spanning the 2021 Acuity Circles transition.

The 2015 lower bound is a scope choice, not a data limit; it concentrates the analysis on the current allocation regime and avoids spending the study on legacy-era preprocessing. The resulting cohort is **494,862 registrations**.

3.3 EXTRACT CONSTRUCTION AND DATA-ENGINEERING PITFALLS

Three properties of the STAR file materially affect correctness and are documented here because each can silently corrupt an analysis:

1. **No header row.** The `.DAT` file carries data from the first line; column names live only in the sibling `.htm` dictionary (475 ordered variables). Column names must be attached from the dictionary, not inferred from the file.
2. **Cohort-defining field.** The waitlisted organ is `WL_ORG` ; the *transplanted* organ `ORGAN` is null until a transplant occurs (missing for 47% of rows — exactly the non-transplant rate). **Filtering the "waitlist" on `ORGAN` silently collapses the cohort to transplant recipients**, producing a spurious ~100%-transplanted population. We filter on `WL_ORG` .
3. **Missing-value sentinel.** STAR encodes missing values as `"."` . Read literally, `"."` defeats null-aware logic (e.g., a still-waiting candidate's blank removal code would not register as missing). We map `"."` to `NA` on ingest.

The extract retains 34 analysis variables plus five derived fields (`outcome` , `event_adverse` , `event_transplant` , `censored` , `days_to_event`) and is written as aggregates/CSV without redistribution outside the team, per the DUA.

3.4 ANALYTIC DATA DICTIONARY

The `KIDPAN` file carries 475 variables; we retain 34 and derive 5. Each column below is tagged by its **role**: *ID* (linkage key, not modeled), *cohort* (used to

define/subset the population), *feature* (a listing-time model input), *outcome* (used to build the label), *leaky* (known only at/after transplant — excluded from models, see §3.7), or *derived*.

Variable	Definition	Role
PT_CODE	Encrypted patient ID; tracks a person across multiple listings	ID
WL_ID_CODE	Encrypted waitlist-registration ID	ID / waitlist filter
TRR_ID_CODE	Encrypted transplant-event ID	ID (transplant-time)
DONOR_ID	Encrypted donor ID	ID (transplant-time)
WL_ORG	Organ the candidate is waitlisted for (KI, KP, PA, PI)	cohort key
ORGAN	Organ actually transplanted (null until transplant)	leaky
WLKI	"Listed for kidney" flag (mostly null in this cohort)	unused
INIT_DATE	Date placed on the waiting list	cohort window + time origin
END_DATE	Date the registration ended (removal/transplant/death/cutoff)	time-to-event
REM_CD	Reason the registration ended (removal code)	primary outcome source
INIT_AGE	Age (years) at listing	feature (numeric)
GENDER	Sex (M/F)	feature (categorical)
ETHCAT	Race/ethnicity category code	feature (categorical)
ABO	Blood group at registration (O, A, B, AB, subtypes)	feature (categorical)

Variable	Definition	Role
INIT_CPRA	Calculated panel-reactive antibody % at listing (sensitization)	feature (numeric)
ON_DIALYSIS	On dialysis at listing (Y/N)	feature (categorical)
BMI_TCR	Body-mass index at registration	feature (numeric)
FUNC_STAT_TCR	Functional-status code at registration (Karnofsky-style)	feature (categorical)
INIT_STAT	Initial medical-urgency status code	feature (categorical)
REGION	OPTN region (1–11)	feature (categorical)
END_CPRA	CPRA at removal / end of episode	leaky (end-of-episode)
END_STAT	Medical-urgency status at removal	leaky (end-of-episode)
DIALYSIS_DATE	Dialysis start date	descriptive (not a feature)
LISTING_CTR_CODE	Listing-center code	descriptive (not a feature)
PREV_TX	Prior-transplant indicator (populated at transplant here)	leaky
DIAG_KI	Primary kidney diagnosis code (populated at transplant here)	leaky
TX_DATE	Transplant date	leaky
DON_TY	Donor type (deceased vs living)	leaky
PTIME	Patient survival time (days)	leaky
PSTATUS	Patient status (1 = dead, 0 = alive)	leaky
COMPOSITE_DEATH_DATE	Best-available death date	leaky

Variable	Definition	Role
<code>DAYSWAIT_CHRON</code>	Total days on the list incl. inactive time	descriptive
<code>DAYSWAIT_ALLOC</code>	Days counted toward allocation priority	descriptive
<code>MULTIORG</code> , <code>A2A2B_ELIGIBILITY</code>	Multi-organ / A2→B eligibility flags (mostly null)	unused
<code>outcome</code>	Coarse outcome bucket derived from <code>REM_CD</code> (§3.6)	derived (label source)
<code>event_adverse</code>	1 if <code>died</code> or <code>removed_too_sick</code> , else 0	derived (classification target)
<code>event_transplant</code>	1 if <code>transplanted</code> , else 0	derived (survival event)
<code>censored</code>	1 if <code>still_waiting</code> at window end	derived (survival censoring)
<code>days_to_event</code>	<code>END_DATE - INIT_DATE</code> in days	derived (survival duration)

3.5 DATA-PREPARATION PIPELINE (RAW → ANALYTIC EXTRACT)

The full pipeline is implemented in `build_analytic_extract.py` (cohort) and `run_analysis.py` (modeling prep). It proceeds in ten steps:

- 1. Attach the schema.** The `.DAT` has no header, so parse the 475 ordered variable names from the sibling `KIDPAN_DATA.htm` dictionary and read the file with `header=None` + those names, under `latin-1` encoding.
- 2. Stream in chunks.** Read the 1.4 GB / 1.3 M-row file in 100 k-row chunks to bound memory; keep only the columns of interest.
- 3. Normalize missingness.** Map the STAR sentinel `"."` to `NA` on read, so null-aware logic behaves.
- 4. Restrict to waitlist registrations.** Keep rows with a registration ID (`WL_ID_CODE` present).

5. **Subset the cohort.** Keep `WL_ORG == "KI"` (kidney-alone) and `INIT_DATE ≥ 2015-01-01`.
6. **Parse dates & durations.** Parse `INIT_DATE / END_DATE`; compute `days_to_event`; null out 59 negative durations (end-before-start data-entry errors).
7. **Derive the outcome.** Map `REM_CD` to the outcome bucket and the `event_* / censored` flags (§3.6).
8. **Project columns.** Retain the 34 analysis variables + 5 derived fields → analytic extract ($494,862 \times 39$), written to CSV with a column manifest and QA summary.
9. **Leakage screen (modeling).** Drop transplant-time/end-of-episode columns tagged *leaky* above, leaving **11 listing-time features** (§3.7).
10. **Preprocess for models.** Numeric features: median-impute + standardize. Categorical features: impute an explicit `MISSING` level + one-hot encode (categories with < 50 occurrences grouped).

The cohort funnel:

Stage	Rows
Raw <code>KIDPAN</code> registrations	1,303,788
After kidney-alone + 2015+ waitlist subset	494,862

3.6 OUTCOME DEFINITION

The primary outcome source is the waitlist removal code `REM_CD`, mapped to coarse buckets (full mapping in Appendix A):

Bucket	<code>REM_CD</code> codes	Interpretation
transplanted	2, 4, 15, 18, 19, 41–45	received a transplant (this registration)
died	8, 21, 23	died on the waiting list

Bucket	REM_CD codes	Interpretation
removed_too_sick	5, 13	removed as too sick to transplant
transplanted_elsewhere	3, 14, 22	transplanted at another center / registration
removed_administrative	7, 9, 10, 11, 16, 17, 20, 24, 40	administrative / non-clinical removal
still_waiting	missing REM_CD	right-censored at window end

The **classification target** is a composite adverse outcome, `event_adverse = 1` if the bucket is `died` or `removed_too_sick`, else 0. The **survival event** is `event_transplant = 1` for `transplanted`, else censored. `days_to_event = END_DATE - INIT_DATE`; 59 rows with negative durations (data-entry errors) were set to missing.

Observed outcome distribution (n = 494,862):

outcome	n	share
transplanted	234,429	47.4%
still_waiting	103,300	20.9%
removed_administrative	45,332	9.2%
transplanted_elsewhere	37,243	7.5%
removed_too_sick	35,219	7.1%
died	34,524	7.0%
unknown	4,815	1.0%
adverse (died + too-sick)	69,743	14.1%

3.7 FEATURE SET AND LEAKAGE CONTROL

Only variables **knowable at the time of listing** are eligible as features. Critically, a cluster of columns in this file is populated **only at transplant** — their missingness (~53%) equals the non-transplant rate — and using them would leak the outcome. **Excluded as leaky:** `DIAG_KI` , `PREV_TX` , `ORGAN` , `DON_TY` , `PSTATUS` , `PTIME` , `TX_DATE` , `DONOR_ID` , `TRR_ID_CODE` , and all `END_*` , `DAYSWAIT_*` , and `COMPOSITE_DEATH_DATE` fields.

The retained **11 listing-time features**:

type	features
numeric	<code>INIT_AGE</code> , <code>INIT_CPRA</code> (calculated PRA at listing), <code>BMI_TCR</code>
categorical	<code>GENDER</code> , <code>ETHCAT</code> (race/ethnicity), <code>ABO</code> (blood type), <code>ON_DIALYSIS</code> , <code>FUNC_STAT_TCR</code> (functional status), <code>INIT_STAT</code> (initial medical-urgency status), <code>REGION</code> (OPTN region), <code>policy_era</code>

`policy_era` is derived from `INIT_DATE` : `AcuityCircles_2021+` if listed on/after 15 March 2021, else `KAS_2015_2021` .

3.8 MISSINGNESS

Feature missingness is low except for sensitization: `INIT_CPRA` 28.8%, `FUNC_STAT_TCR` 0.8%, `BMI_TCR` 0.3%; the remaining features are effectively complete. Numeric features were median-imputed; categorical features used an explicit `MISSING` level so that "not recorded" is itself a modelable signal rather than silently dropped.

4. Methods

4.1 PROBLEM FORMULATION

- **Classification (Q1).** Predict `event_adverse` from listing-time features. Positive prevalence 14.1% (imbalanced), so PR-AUC and calibration lead the evaluation, with ROC-AUC reported for comparability.
- **Survival (Q2).** Model time from listing to transplant. Non-transplant terminations (death, removal, still-waiting) are treated as right-censoring; this yields a **cause-specific** time-to-transplant hazard, with the competing-risks caveat discussed in Section 6.

4.2 PREPROCESSING

A single `ColumnTransformer` applied median imputation + standardization to numeric features and most-frequent imputation + one-hot encoding (rare levels < 50 grouped) to categoricals. The same transformer definition was reused across models to keep comparisons clean.

4.3 CLASSIFICATION MODELS

- **Logistic regression** (baseline): L2-regularized, `class_weight="balanced"`, 2,000 max iterations.
- **XGBoost** [16]: 400 trees, depth 5, learning rate 0.05, subsample 0.8, colsample 0.8, `scale_pos_weight` set to the negative/positive ratio, objective tuned to `aucpr`.

Split. Stratified 75/25 train/test. The training portion was further split 80/20 into a model-fit set and a **calibration hold-out** (Section 4.4). The test set was never used for fitting or calibration.

4.4 POST-HOC CALIBRATION

Class-imbalance weighting (`scale_pos_weight`) improves minority-class recall but **inflates predicted probabilities**, harming the Brier score and reliability — the calibration failure mode emphasized by Van Calster *et al.* [12], [13]. We therefore fit an **isotonic regression** on the calibration hold-out mapping raw XGBoost scores to empirical frequencies, and applied it to the test set. Because isotonic regression is **monotonic**, it leaves rank-order metrics (ROC-AUC, PR-AUC, lift) unchanged while correcting probability magnitudes.

4.5 EVALUATION METRICS

ROC-AUC; PR-AUC (baseline = prevalence); Brier score (lower better); lift in the top risk decile; and, for survival, **Harrell's C-index** [18]. Calibration is shown as reliability curves (quantile-binned).

4.6 FAIRNESS AUDIT

For the final XGBoost model we computed **subgroup ROC-AUC** across sex, race/ethnicity, blood type, OPTN region, and age band, reporting each subgroup's gap versus the overall test AUC. The success tolerance is $|\text{gap}| \leq 0.05$. Subgroups with < 500 members or a single outcome class were suppressed.

4.7 EXPLAINABILITY

Global attribution used **SHAP** [17] via XGBoost's native `pred_contribs` on a 4,000-row test sample, summarized as mean $|\text{SHAP}|$ (average impact on the log-odds of an adverse outcome).

4.8 ASSUMPTIONS (EXPLICIT)

We state assumptions plainly so reviewers can weigh them:

1. **Registration is the unit of analysis**, not the patient. A patient listed at multiple centers contributes multiple rows; we did not de-duplicate to the patient level, so estimates are per-registration.

2. **REM_CD** mapping is a working hypothesis. The bucket assignments (Appendix A) approximate the dictionary's REMCD format; multi-listing and administrative codes in particular warrant validation before clinical use.
3. **transplanted_elsewhere** is treated as a non-adverse, non-event outcome. It is neither an observed failure of this registration nor a transplant we allocated; it is effectively an informative removal, which we do not model as a competing event here.
4. **Right-censoring is assumed non-informative** for the survival model — i.e., censoring is unrelated to the transplant hazard given covariates. This is imperfect (sicker patients are removed), motivating the competing-risks extension.
5. **The classifier's negative label includes still-waiting candidates**, some of whom will later experience an adverse event. This biases the classifier conservatively; the survival model is the principled complement.
6. **Cross-sectional listing features are static**. We use values recorded at registration and do not incorporate time-varying updates (e.g., changing CPRA, status, or dialysis vintage).
7. **policy_era** is a proxy for allocation regime by listing date; it does not capture within-era policy nuance or local OPO behavior.
8. **Random train/test split** assumes exchangeability across the period; a temporal split (train pre-2021, test post-2021) is the stricter stress test and is proposed as future work.
9. **The 150k sub-sample for Cox** is assumed representative of the full cohort for C-index estimation; it was drawn at random with a fixed seed.
10. **Missing-as-signal**: encoding **MISSING** as a category assumes missingness carries information (often true in registries), which can introduce site-specific artifacts.

4.9 REPRODUCIBILITY AND IMPLEMENTATION

Python 3.12 in an isolated virtual environment. Libraries: pandas 2.3, scikit-learn 1.9, XGBoost 3.4, lifelines 0.30, SHAP 0.49, matplotlib. All randomness fixed (`random_state = 42`). Two scripts reproduce the study end-to-end:

`build_analytic_extract.py` (cohort construction from the STAR file) and `run_analysis.py` (EDA → models → fairness → SHAP → report). **Only aggregates and figures are emitted; no row-level records are written or shared**, consistent with the DUA.

5. Results

5.1 COHORT AND OUTCOMES

The cohort of 494,862 kidney-alone registrations (Section 3.6) shows 47.4% transplanted, 20.9% still waiting at window end, and 14.1% experiencing the adverse composite (death or too-sick removal). Adverse rate rises with age and differs by policy era (Figure `adverse_rate_breakdown.png`).

5.2 CLASSIFICATION — WHO IS AT RISK?

model	ROC-AUC	PR-AUC	Brier	Lift@top-decile
Logistic Regression	0.716	0.275	0.218	2.35x
XGBoost (isotonic-calibrated)	0.726	0.284	0.111	2.51x

Test-set prevalence 14.1% (PR-AUC baseline). Against the aspirational success targets:

metric	target	achieved (XGBoost)	met?
ROC-AUC	≥ 0.78	0.726	✗
PR-AUC	≥ 0.65	0.284	✗
Brier	≤ 0.18	0.111	✓
Lift@decile	≥ 2.5	2.51x	✓

The top-decile lift of $2.51\times$ means the model's highest-risk 10% of candidates experience adverse outcomes at $\sim 2.5\times$ the base rate — an operationally useful concentration for prioritizing supportive care or review, even where absolute discrimination is modest. (Figures: [roc_pr_curves.png](#) , [decile_lift.png](#) .)

5.3 CALIBRATION

Isotonic calibration reduced the XGBoost Brier score from **0.212 (raw)** to **0.111**, moving it from failing to comfortably within the ≤ 0.18 target. The reliability curve ([calibration.png](#)) shows the raw model systematically over-predicting risk (a direct consequence of [scale_pos_weight](#)), with the calibrated curve tracking the diagonal. Rank metrics were unchanged, confirming the calibration was a pure probability-magnitude correction.

5.4 SURVIVAL — HOW LONG UNTIL TRANSPLANT?

The Cox proportional-hazards model achieved a **C-index of 0.629** for time-to-transplant (below the 0.72 target). Among transplanted candidates, **median observed days-to-transplant ranged 211–400 across OPTN regions** — a nearly two-fold geographic spread. Kaplan–Meier curves by policy era ([km_survival.png](#)) visualize the shift in the "not-yet-transplanted" probability over time.

5.5 FAIRNESS AUDIT

Overall test AUC 0.726. All audited dimensions stayed within the 0.05 subgroup-AUC tolerance:

dimension	worst subgroup (AUC)	best subgroup (AUC)	max gap	within 0.05?
Sex	M/F $\approx$ 0.72–0.73	—	0.004	✓
Race/ethnicity	Black 0.703	Hispanic 0.749	0.024	✓
Blood type	A1 0.683	O 0.731	0.044	✓
OPTN region	Region 3 = 0.703	Region 6 = 0.761	0.034	✓
Age band	35–49 = 0.685	< 35 = 0.719	0.042	✓

Two caveats temper the green checks. First, the largest gaps (blood type A1, older age bands) approach the tolerance and rest on smaller subgroups. Second — and more important — **AUC parity is a statement about ranking within a subgroup, not about equal outcomes across subgroups**. The 211–400-day regional wait-time spread (Section 5.4) is a substantive equity concern that AUC parity does not capture (Section 6). (Figure: [fairness_region_auc.png](#).)

5.6 EXPLAINABILITY

The dominant drivers of predicted adverse risk (mean $|\text{SHAP}|$) were, in order: **functional status at listing** ([FUNC_STAT_TCR](#) , by a wide margin), **policy era** (Acuity Circles), **BMI**, **race/ethnicity**, **OPTN region**, and **blood type** ([shap_importance.png](#)). The primacy of functional status is clinically coherent — impaired functional status is an established predictor of waitlist mortality — and reassuring in that the top driver is a clinical variable rather than a purely administrative one. Region and policy-era prominence, however, indicate the model is partly learning **system and geography**, which is the crux of the fairness discussion below.

FIGURES

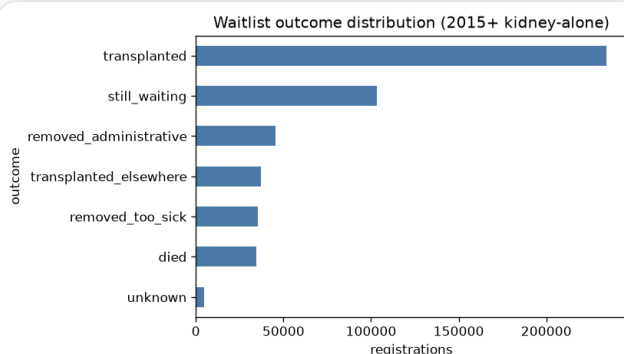


Fig 1 Waitlist outcome distribution — 2015+ kidney-alone cohort (n = 494,862).

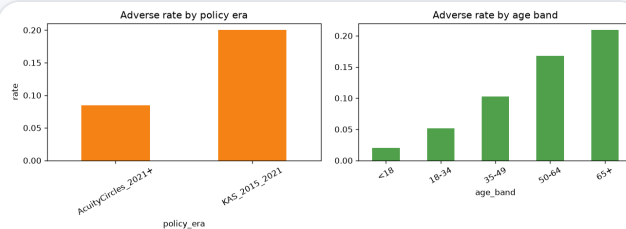


Fig 2 Adverse-outcome rate by policy era and by age band.

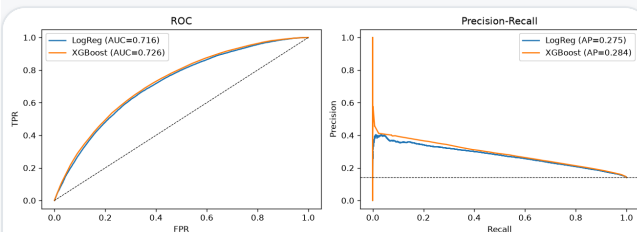


Fig 3 ROC and Precision–Recall curves: logistic-regression baseline vs. XGBoost.

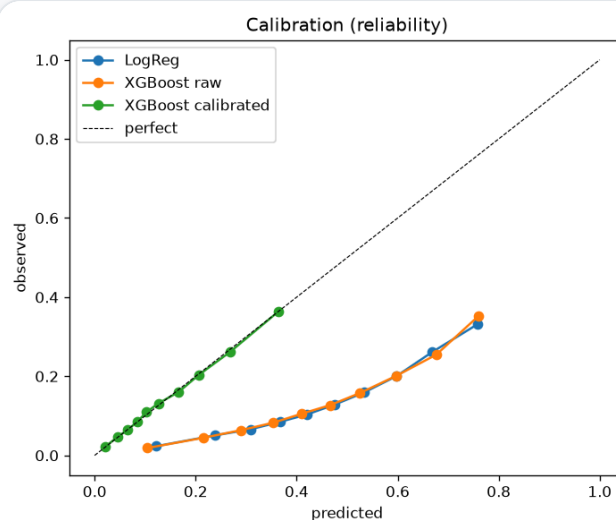


Fig 4 Reliability curves. Raw XGBoost systematically over-predicts; isotonic calibration restores the diagonal.

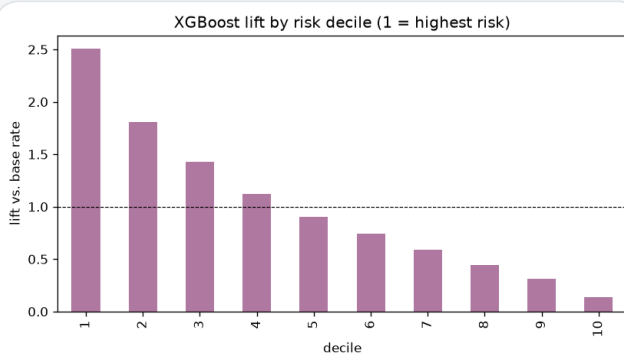


Fig 5 Lift by predicted-risk decile (decile 1 = highest risk). Top decile $\approx 2.5\times$ base rate.

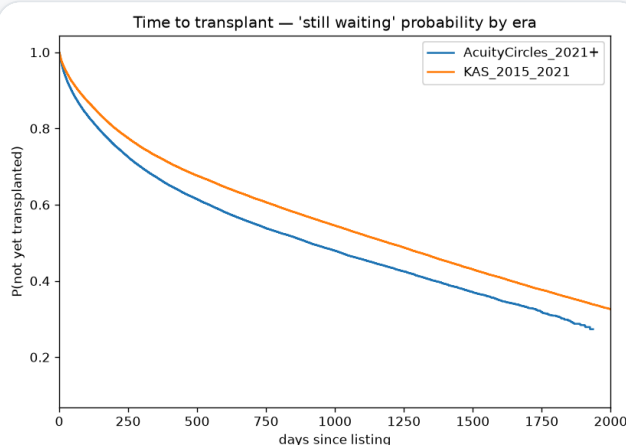


Fig 6 Kaplan–Meier time-to-transplant by policy era — probability of not-yet-transplanted.

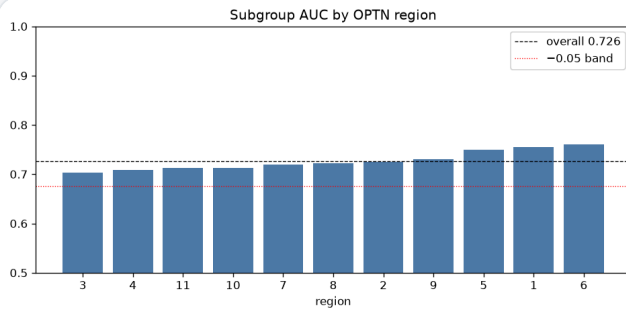


Fig 7 Subgroup AUC by OPTN region against the overall AUC and the -0.05 tolerance band.

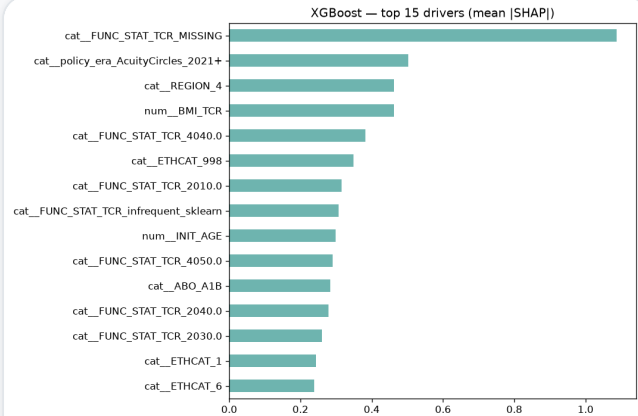


Fig 8 Top model drivers by mean ISHAPI — functional status at listing dominates.

6. Discussion

Why performance falls short of the aspirational AUC targets — and why that is informative. With only 11 listing-time features, the model cannot see the strongest determinants of both risk and wait time: HLA matching and sensitization dynamics, KDPI of available donors, dialysis vintage, dynamic CPRA, local OPO behavior, and comorbidity burden beyond a single functional-status code. The registry contains richer signals, but many are transplant-time (and thus leaky) or absent from this extract. The honest reading is that **waitlist risk is only moderately predictable from listing-time demographics and coarse clinical status**; a complex model that barely outperforms logistic regression (0.726 vs 0.716 ROC-AUC) is itself a finding — most of the separable signal is captured by simple main effects.

Calibration matters more than the AUC gap for deployment. A model used to *prioritize* candidates needs trustworthy probabilities, not just correct ordering. The $0.212 \rightarrow 0.111$ Brier improvement is the difference between a model that says "everyone is high-risk" and one whose stated probabilities match observed frequencies. We argue calibration should be a required deliverable, not an afterthought, whenever class-imbalance weighting is used.

Fairness: ranking parity vs. allocation equity. The subgroup-AUC audit passes, but the more consequential disparity is in *access*: median wait varies up to two-fold by region, and region/policy-era rank among the model's top drivers. A model that predicts risk well within every subgroup can still encode and perpetuate an inequitable allocation geography — the same failure mode Obermeyer *et al.* [9] documented when an algorithm optimized a biased proxy, and the same access inequity that motivated race-free eGFR reform [10], [11]. We therefore recommend reporting **absolute outcome disparities alongside AUC parity**, and treating geographic drivers with suspicion in any deployed risk score.

Competing risks. Treating death/removal as censoring for the time-to-transplant model overstates transplant probability, because a candidate who dies can never be transplanted. The cause-specific hazard we estimate answers "among those still at risk, what is the transplant rate," not "what is a candidate's probability of transplant." As Sapir-Pichhadze *et al.* [8] show for exactly this population, a Fine–Gray subdistribution model [15] (or a cause-specific competing-risks framework) is the correct next step and is expected to change wait-time estimates.

7. Limitations

- **Per-registration, not per-patient:** multi-listing inflates counts and can double-count patients (Assumption 1).
- **Static features:** no time-varying covariates (CPRA, status, dialysis vintage evolve during waiting).
- **REM_CD mapping unvalidated** against every code; `unknown` (1.0%) and `transplanted_elsewhere` handling are approximations.
- **Informative censoring:** sicker patients are removed, violating the non-informative-censoring assumption of the basic Cox model.

- **No temporal validation:** a random split may overstate stability across the 2021 policy change.
- **Single extract / single release:** results are specific to the 202606 STAR release and the 2015+ window.
- **Free-text fields** (cause-of-death narratives, comorbidity notes) were out of scope; they may carry additional signal.
- **Aggregate-only outputs** preclude external row-level replication (a deliberate DUA constraint, not a scientific choice).

8. Ethical Considerations and Data Governance

The STAR data are governed by an OPTN Data Use Agreement. This study wrote **only aggregate statistics and figures**; no row-level records were exported, committed, or redistributed. Encrypted patient/registration identifiers were used solely for cohort construction and never published. Any deployment of a waitlist-risk score raises the central ethical risk that the model **re-learns historical allocation patterns as if they were clinical truth**; the fairness audit and the explicit surfacing of geographic drivers are intended to keep that risk visible. A risk score of this kind should inform supportive-care prioritization and candidate counseling, **not** ration organs, and should never be deployed without prospective, subgroup-stratified validation and clinical governance.

9. Next Steps

1. **Competing-risks survival** (Fine–Gray / cause-specific) and calibrated time-to-event predictions, replacing the censoring approximation.

2. **Temporal validation:** train on KAS-era listings, test post-Acuity-Circles, to quantify distribution-shift robustness.
3. **Richer features:** incorporate sensitization dynamics, dialysis vintage, KDPI exposure, and comorbidity indices where available at listing.
4. **Absolute-disparity fairness metrics** (wait-time gaps, transplant-rate ratios) reported alongside AUC parity; consider fairness-constrained or reweighted training.
5. **Time-varying models** (landmarking or joint models) to use updates during waiting.
6. **Patient-level de-duplication** to convert per-registration to per-patient estimates.
7. **Prospective, governed pilot** with clinician-in-the-loop evaluation before any operational use.
8. **Decision-curve / operating-point analysis** tied to a concrete use case (e.g., which decile triggers palliative-care referral).
9. **Interactive delivery:** a Streamlit application exposing calibrated risk, expected-wait curves, and per-candidate SHAP explanations for review (see deployment note).

10. Conclusion

On national kidney-waitlist data, listing-time features predict adverse waitlist outcomes and time-to-transplant to a moderate, clinically suggestive degree (ROC-AUC 0.726; C-index 0.629), with a top-decile lift ($2.51\times$) and calibrated probabilities (Brier 0.111) that are operationally useful for prioritization. The study's most transferable lessons are methodological: define the waitlist cohort on the *waitlisted* organ, guard against transplant-time leakage, calibrate probabilities whenever imbalance weighting is used, and report absolute outcome disparities —

especially geographic — alongside subgroup-AUC parity. The gap between the model's performance and the aspirational targets is not a failure but a measured statement of how much of waitlist risk is legible from listing-time registry data alone.

Appendix A — REM_CD → outcome mapping

bucket	REM_CD codes
transplanted	2, 4, 15, 18, 19, 41, 42, 43, 44, 45
died	8, 21, 23
removed_too_sick	5, 13
transplanted_elsewhere	3, 14, 22
removed_administrative	7, 9, 10, 11, 16, 17, 20, 24, 40
still_waiting	REM_CD missing (right-censored)
unknown	any code not listed above

Appendix B — Reproducibility

- **Extract:** `build_analytic_extract.py --input KIDPAN_DATA.DAT --outdir extract_2015_kidney_only --min-year 2015 (cohort WL_ORG == "KI" )`.
- **Analysis:** `run_analysis.py → REPORT.md + figures/`.
- **Environment:** Python 3.12; pandas 2.3, scikit-learn 1.9, XGBoost 3.4, lifelines 0.30, SHAP 0.49, matplotlib. Seed 42.

- **Figures:** `outcome_distribution.png` , `adverse_rate_breakdown.png` , `roc_pr_curves.png` , `calibration.png` , `decile_lift.png` , `km_survival.png` , `fairness_region_auc.png` , `shap_importance.png` .

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Aggregates and figures only — no row-level records, consistent with the OPTN Data Use Agreement. Reproducible from `build_analytic_extract.py` and `run_analysis.py` on the 202606 STAR release.